N100 FINANCIAL INTELLIGENCE PLATFORM

Final Project Report

Bluestock Fintech Capstone Project

1. Executive Summary

The N100 Financial Intelligence Platform is an automated financial analytics system developed to analyze Nifty 100 companies. The platform combines data engineering, financial analysis, valuation models, dashboard visualization, NLP-based insights, cash flow analysis, and automated reporting.

The goal of this project is to provide investors and analysts with a single platform for evaluating company performance, financial strength, valuation, and business quality.

2. Project Overview

Problem

Analyzing multiple companies manually requires large amounts of time and effort. Investors need financial metrics, comparison tools, valuation analysis, and business insights in one place.

Solution

The N100 Financial Intelligence Platform automates the complete analysis process by:

- Processing financial datasets
- Calculating financial ratios
- Comparing companies
- Screening stocks
- Performing valuation analysis

- Generating automated reports

3. Objectives

The objectives of this project are:

- Build an automated financial data pipeline
- Create a structured financial database
- Calculate important company metrics
- Develop an interactive dashboard
- Build screening and peer comparison systems
- Generate valuation insights
- Create automated financial reports

4. Technology Stack

- Programming: Python
- Data Processing: Pandas, NumPy
- Database: SQLite
- Dashboard: Streamlit
- Visualization: Plotly
- Reports: ReportLab
- Data Format: CSV, Excel

5. System Architecture

Data Layer

Input datasets:

- Financial Ratios
- Profit & Loss Statements
- Balance Sheet Data
- Cash Flow Data
- Market Capital Data
- Sector Information

Processing Layer

Activities:

- Data cleaning
- Data transformation
- Feature engineering
- Financial calculations

Intelligence Layer

Modules:

- NLP Pros & Cons Generator
- Cash Flow Intelligence
- Capital Allocation Analysis
- Valuation Engine

Output Layer

Generated:

- Dashboard
- Excel reports
- CSV outputs
- PDF reports

6. Implemented Modules

6.1 ETL Pipeline

The ETL pipeline prepares raw financial data by:

- Removing inconsistencies
- Handling missing values
- Creating structured datasets
- Loading data into database

6.2 Financial Ratio Engine

Calculated metrics:

- ROE
- ROCE
- Net Profit Margin
- Operating Profit Margin
- Debt-to-Equity Ratio
- Interest Coverage Ratio
- Free Cash Flow

6.3 Screener Module

- Features:
- Quality screening
- Value screening
- Growth screening
- Dividend screening
- Debt-free screening
- Turnaround screening

6.4 Peer Comparison Module

Companies are compared using:

- Profitability
- Growth
- Debt levels
- Valuation metrics
- Visualization:
- Radar charts
- KPI comparison tables

6.5 Valuation Module

Calculated:

- P/E Ratio
- P/B Ratio
- EV/EBITDA
- FCF Yield

Classification:

- Discount

- Fair Value
- Caution

6.6 NLP Intelligence Module

Automatically generates:

Positive Insights

- Strong ROE
- Healthy cash flow
- Low debt
- Strong growth

Negative Insights

- High leverage
- Weak margins
- Profitability concerns

Output:

- pros_cons_generated.csv

6.7 Cash Flow Intelligence

Analyzes:

- CFO Quality Score
- CapEx Intensity
- Distress Signals
- Deleveraging Patterns
- Capital Allocation

Output:

- cashflow_intelligence.xlsx

7. Streamlit Dashboard

The dashboard contains 8 screens:

- Home Dashboard

- Company Profile
- Screener
- Peer Comparison
- Trend Analysis
- Sector Analysis
- Capital Allocation
- Reports

8. Project Outputs

Generated files:

- pros_cons_generated.csv
- cashflow_intelligence.xlsx
- valuation_summary.xlsx
- valuation_flags.csv
- company tearsheet PDFs
- sector reports
- portfolio reports

9. Results

The platform successfully analyzes:

- 92 Nifty 100 companies
- Multiple years of financial data
- Company performance metrics
- Valuation indicators
- Cash flow quality
- Investment insights

10. Challenges

- Challenges faced:
- Cleaning inconsistent company names

- Handling missing financial values
- Integrating multiple datasets
- Designing dashboard components
- Creating automated PDF layouts

Solutions:

- Data validation
- Error handling
- Modular programming approach

11. Future Scope

Future improvements:

- Real-time stock mark
- Machine learning predictions
- Portfolio optimization
- News sentiment analysis
- AI investment assistant
- Automated email reports

12. Conclusion

The N100 Financial Intelligence Platform successfully combines financial analytics, automation, visualization, and reporting into a single solution.

The platform reduces manual analysis effort and provides investors with structured insights for better financial decision-making.